

STATISTICS

Overview of Statistics

The curriculum in liberal arts and sciences with a major in statistics is designed to prepare students for (1) statistics positions in business, industry or commerce, nonprofit institutions, and in state or federal government; and (2) graduate study in statistics. Positions include the following types of work: statistical design, data visualization, analysis and interpretation of experiments and surveys; data processing and analysis using modern computation facilities and statistical computing systems; application of statistical principles and methods in commercial areas such as finance, insurance, industrial research, technology, marketing, manufacturing, sports analytics, quality control, and nonprofit organizations, such as institutions involved in health care or medical research.

Given the application of statistical work to a broad range of fields, students majoring in statistics often obtain at least a minor in a field of application. Some common minors earned by statistics majors are Economics (<https://catalog.iastate.edu/collegeofliberalartsandsciences/economics/#undergraduateminortext>), General Business (<https://catalog.iastate.edu/collegeofbusiness/>), and Mathematics (<https://catalog.iastate.edu/collegeofliberalartsandsciences/mathematics/#undergraduateminortext>). Students preparing for positions in data analysis should consider a minor or certificate in Data Science (<https://catalog.iastate.edu/collegeofliberalartsandsciences/datascience/#overviewtext>). Students preparing for a career in the field of actuarial science should consider a certificate in Actuarial Science (<https://catalog.iastate.edu/collegeofbusiness/actuarialscience/#certificatetext>). Students intending to pursue graduate study in Statistics are strongly advised to complete at least a minor in Mathematics (<https://catalog.iastate.edu/collegeofliberalartsandsciences/mathematics/#undergraduateminortext>) including credit in MATH 4140 Analysis I.

Many Statistics majors earn an additional major or degree in a field of application or in Mathematics. Your academic advisor can assist you in developing your course of study including other majors.

Student Learning Outcomes

Students completing the undergraduate degree in statistics should have a broad understanding of the discipline of statistics. Upon graduation, students should be able to:

- Design observational studies and experiments in order to efficiently collect data to help answer questions about science, technology, and society.
- Analyze data arising from observational studies and experiments in order to help answer questions about the world around them.

- Use modern statistical computing to aid in the collection and analysis of data.
- Explain and apply the mathematical and theoretical basis for probability and statistical inference to help answer questions about the world around them.
- Effectively communicate statistical findings using oral, visual and written formats.

Undergraduate Major

The requirements for the undergraduate major in statistics are:

STAT 1100	Orientation in Mathematics and Statistics	1
STAT 2010	Introduction to Statistical Concepts and Methods	4
STAT 3201	Intermediate Statistical Concepts and Methods	4
STAT 3441	Introduction to the Theory of Probability and Statistics I	4
STAT 3442	Introduction to the Theory of Probability and Statistics II	4
STAT 4207	Multivariate Data Analysis	3
STAT 4212	Experimental Design and Data Analysis	3
STAT 4278	Computer Processing of Scientific Data	3
STAT 4279	Statistical Computing for Data Analysis	3
A minimum of 6 credits from the following:		6
STAT 3610	Statistical Quality Assurance	
STAT 4204	Regression for Social and Behavioral Research	
STAT 4206	Spatial Data Analysis	
STAT 4215	Time Series Data Analysis	
STAT 4217	Categorical Data Analysis	
STAT 4219	Bayesian Data Analysis	
STAT 4221	Survey Sampling Design and Data Analysis	
STAT 4232	Empirical Methods for the Computational Sciences	
STAT 4237X	Sports Analytics	
Completion of one of the following options:		12
Option I		
MATH 1650	Calculus I	
MATH 1660	Calculus II	
MATH 2650	Calculus III	
Option II		
MATH 1650	Calculus I	
MATH 1660H	Calculus II: Honors	
MATH 2650H	Calculus III: Honors	
Credit in one of the following:		3-4
MATH 2070	Matrices and Linear Algebra	
MATH 3170	Theory of Linear Algebra	

Credit in one of the following:	3
COMS 1270 Introduction to Computer Programming	
COMS 2070 Fundamentals of Computer Programming	
DS 2010 Introduction to Data Science	
DS 2020 Data Acquisition and Exploratory Data Analysis	
Credit in one of the following:	3
COMST 2110 Interpersonal Communication	
SPCM 2120 Fundamentals of Public Speaking	
Credit in one of the following (with a grade of C- or higher):	3
ENGL 3020 Business Communication	
ENGL 3020H Business Communication: Honors	
ENGL 3140 Technical Communication	
ENGL 3140H Technical Communication: Honors	
Total Credits	59-60

As majors in the College of Liberal Arts and Sciences, Statistics students must meet College of Liberal Arts and Sciences (<https://catalog.iastate.edu/collegeofliberalartsandsciences/#lascollegerequirementstext>) and University-wide requirements (<https://catalog.iastate.edu/collegescurricula/>) for graduation in addition to those stated above for the major.

LAS majors require a minimum of 120 credits, including a minimum of 45 credits at the 3000/4000 level or above taken at a four-year college. All such credits, including courses taken on a pass/not pass basis, may be used to meet this requirement. You must also complete the LAS world language requirement and career proficiency requirement.

Students in all ISU majors must complete a three-credit course in U.S. Cultures and Communities and a three-credit course in International Perspectives. Check (<http://www.registrar.iastate.edu/courses/div-ip-guide.html>) for a list of approved courses. Discuss with your advisor how the two courses you select can be applied to your academic plan.

Statistics, B.S.

Below is a suggested pathway for new majors. This plan is an example only; students should consult their advisers for their individualized needs.

Freshman

Fall	Credits Spring	Credits
ENGL 1500	3 MATH 1660 or 1660H	4
LIB 1600	1 STAT 3201	4
STAT 1100	1 Computer Science or Data Science	3
MATH 1650	4 Natural Science	4
STAT 2010	4	

Arts & Humanities	3
16	15

Sophomore

Fall	Credits Spring	Credits
MATH 2650 or 2650H	4 MATH 2070 or 3170	3-4
STAT 4212	3 STAT 4279	3
ENGL 2500	3 SPCM 2120 or COMST 2110	3
Social Science	3 Social Science	3
LAS 2030	1 Arts & Humanities	3
	14	15-16

Junior

Fall	Credits Spring	Credits
STAT 3441	4 STAT 3442	4
STAT 4278	3 ENGL 3140 or 3020	3
Arts & Humanities	3 Social Science	3
World Language/Elective	3-4 World Language/Elective	3-4
Elective	3 Elective	3
16-17		16-17

Senior

Fall	Credits Spring	Credits
STAT 4207	3 Statistics 3000+ Elective	3
Statistics 3000+ Elective	3 Arts & Humanities	3
Natural Science	4 3000+ Electives	6
3000+ Electives	6	
	16	12

Total Credits: 120-123

Undergraduate Minor

The department offers a minor in statistics which may be earned by satisfying the following requirements:

- Credit in one introductory level course in statistics, chosen from the following list: STAT 1010, STAT 1040, STAT 2010, STAT 2026, STAT 3031, STAT 3005, STAT 3022, STAT 3030.
- Credit in one intermediate level course in statistics, chosen from the following list: STAT 3201, STAT 3226.
- At least 9 additional credits in statistics courses, chosen from the following list: STAT 3441, STAT 3442, STAT 3447, STAT 3610, STAT 4204, STAT 4206, STAT 4212, STAT 4215, STAT 4217, STAT 4219, STAT 4221, STAT 4232, STAT 4237X, STAT 4278, STAT 4279.

At least 3 credits used to complete the minor may not be used to meet any other department, college, or university requirement.

All credits used to meet the minor requirements may also be used to satisfy the credit requirement for graduation and to meet credit requirements in courses numbered 3000 or above.

Courses used to complete the minor may not be taken on a pass-not pass basis.

The Statistics Undergraduate Minor is an LAS Minor. In addition to University policies governing minors (<https://catalog.iastate.edu/academics/#degreeplanningtext>), LAS minors require at least 6 credits in courses numbered 3000 and above, with a grade of C or higher.

Graduate Study

The department offers graduate programs leading to the Master of Applied Statistics degree, and the Master of Science (M.S.) and the Doctor of Philosophy (Ph.D.) degrees with a major in Statistics.

Graduates of the Master of Applied Statistics program develop a strong foundation in statistical applications, learning to correctly use and interpret the most frequently employed classical and modern methods in statistics. The students will gain skills in selecting appropriate techniques for data-driven problems, explaining theoretical foundations of classical and modern statistical methods, and effectively communicating statistical results to all stakeholders. Students will also learn to use modern statistical computing technologies, and to apply machine learning and artificial intelligence to enhance insights from data. Students complete a minimum of 30 semester credits, including statistical consulting coursework and credit for statistical work experience through an internship or current employer.

Graduates of the M.S. program have studied basic statistical theory and methods. Elective courses in the M.S. program provide an opportunity for students to emphasize particular areas of application in their program. Students complete a minimum of 34 semester credits, including work on a capstone project resulting in a written creative component under the direction of an individual major professor and defended in a final oral examination.

Graduates of the Ph.D. program have demonstrated mastery of basic statistical theory and methods, studied advanced statistical theory and methods, and have demonstrated the ability to conduct independent research resulting in an original contribution to the discipline. Candidates for the Ph.D. degree in Statistics complete a minimum of 72 semester credits, including at least 18 credits given for research activity, pass an oral preliminary examination, and submit a written dissertation containing original research that is defended in a final oral examination. Dissertation research is typically conducted in close collaboration with a major professor and usually results in publishable material. The

department does not offer specific program tracks or areas of emphasis, but the diversity of elective courses and research areas of faculty allow students to tailor their individual programs to reflect areas of particular interest.

The Ph.D. degree is also offered as a co-major with other graduate programs. Graduates of co-major Ph.D. programs in Statistics and an applied scientific discipline have demonstrated mastery of basic statistical theory and methods and have studied advanced statistical methodology. Students complete a minimum of 72 semester credits for courses in statistics and the chosen scientific discipline. Students conduct research that is a combination of statistical methodology and the scientific discipline. Co-major professors work with the student to prepare for an oral preliminary examination and to conduct research leading to a single dissertation project, defended in a final oral examination, that produces an original contribution to at least one of the two disciplines.

Graduates of co-major Ph.D. programs in statistics and an area of theoretical mathematics have demonstrated mastery of basic statistical theory and methods and have studied advanced statistical theory. Students complete a minimum of 72 semester credits for courses in statistics and the other discipline. Co-major professors work with the student to prepare for an oral preliminary examination and to conduct research leading to a single dissertation project, defended in a final oral examination, that makes an original contribution at the interface of statistical theory and a sub-discipline of mathematics.

Graduate work leading to a minor in statistics is available for students majoring in other programs, at both the M.S. and Ph.D. levels.

The department participates in inter-disciplinary graduate programs in Bioinformatics and Computational Biology, Ecology and Evolutionary Biology, Genetics, Human Computer Interaction, Nutritional Sciences, and Wind Energy Science, Engineering, and Policy.